DATA PORTFOLIO PROJECT REPORT

Airline Delay & Performance Analysis Dashboard

An end-to-end operational framework explaining schedule patterns using Python, SQL, and Power BI

PREPARED BY

Sandleen Sethi

DOMAIN

Aviation Business Intelligence

TOOLS USED

Python | SQL | Power BI

PROJECT SCOPE

Portfolio Analytics Case Study

TABLE OF CONTENTS

This document is organized into distinct sections to show the process of cleaning, analyzing, and visualizing airline records. Every main topic begins on a fresh page.

CHAPTER 1: INTRODUCTION & MAIN PROJECT GOALS

Flight delays represent a significant driver of operational deficits for commercial carriers and directly affect customer satisfaction metrics. Identifying the root causes behind schedule variances is paramount to optimizing network flight schedules. This report audits a massive historical dataset to isolate structural bottlenecks and evaluate system-wide operational efficiency.

1.1 Core Metric Baselines

System-Wide Schedule Variance Indicators:

- **Total System Delayed Volume: 2.0 Million flights** across the network suffered deviations from scheduled departure or arrival times.
- **Average Disruption Duration:** The global network recorded a precise **average delay duration of 9.77 minutes** per affected flight, serving as a critical operational benchmark.

1.2 Project Objectives

The main strategic goals of this initiative are broken down clearly below:

1. Clean Raw Aviation Files

Fix missing values and prepare large datasets so that the numbers are mathematically accurate and ready for operational analysis.

2. Calculate Clear Core Metrics

Use database code to find flight totals, isolate average delay intervals, and establish performance rankings across different weekdays.

3. Design Interactive Dashboards

Build easy-to-use visual pages in Power BI so that decision-makers can filter results by airline, date, or airport instantly.

CHAPTER 2: DATA MODEL & TABLE CONNECTIONS

To optimize dashboard query speeds and maximize responsiveness, the data tables were linked using a Star Schema layout. This structure connects a high-density central transaction table directly to low-density, descriptive dimension lookups.

2.1 Visual Database Schema Layout

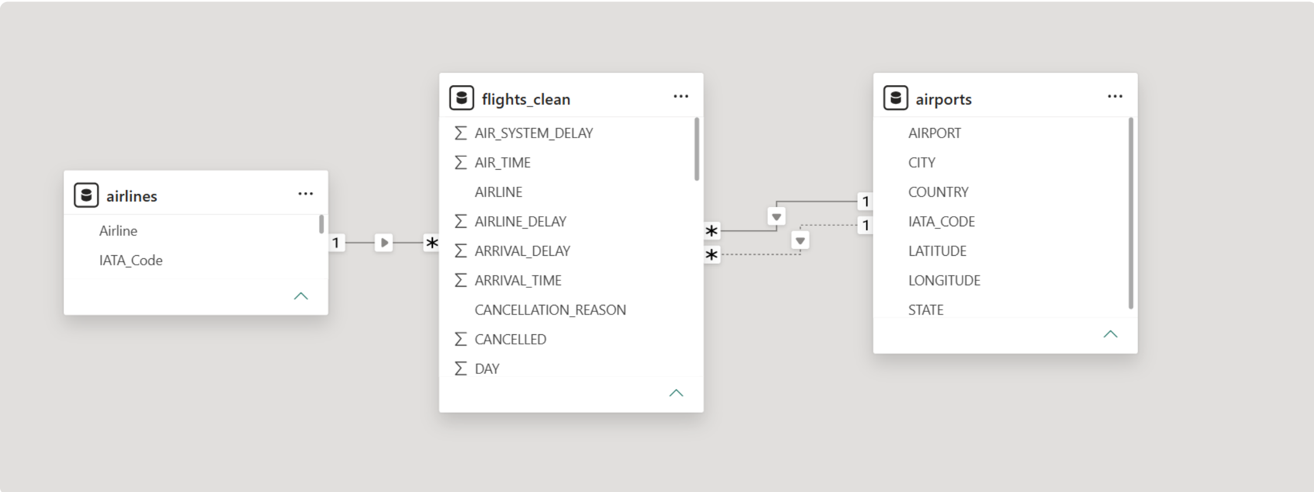


Figure 2.1: The Power BI relationship layout depicting how lookup dimension tables filter down into the central transaction fact table.

2.2 Main Table Explanations

1. Core Fact Table (`flights_clean`)

The primary central ledger containing over 5 million rows. It stores the transactional numbers for every recorded flight vector, including actual air time, departure delay, and arrival delay minutes.

2. Airlines Lookup Table (`airlines`)

A normalized lookup table translating two-character IATA carrier codes into full corporate identities (like translating "AA" to "American Airlines").

3. Airports Lookup Table (`airports`)

A descriptive geospatial dimension list tracking airport locations, containing full city names, states, latitudes, and longitudes.

CHAPTER 3: CLEANING THE DATA WITH PYTHON

Before designing dashboard elements, raw data files were programmatically audited via Python to clean data anomalies. The missing row patterns were not system errors; instead, they showed standard on-time operations where no delay occurred.

3.1 Missing Values and Fixes Matrix

Column Name	Missing Rows	Missing %	Programmatic Remediation Strategy
CANCELLATION_REASON	5,729,195	98.45%	Retained unaltered as a valuable categorical text field for canceled flight subsets.
LATE_AIRCRAFT_DELAY	4,755,640	81.72%	Imputed with 0 (indicating nominal, on-time aircraft turnarounds).
AIR_SYSTEM_DELAY	4,755,640	81.72%	Imputed with 0 (indicating optimal air traffic system control baseline).
WEATHER_DELAY	4,755,640	81.72%	Imputed with 0 (indicating absence of adverse meteorological impacts).
SCHEDULED_TIME	6	<0.001%	Removed these 6 rows from the file completely.

3.2 Simple Python Cleaning Script

```
# Fill missing delay category counts with zero values
delay_columns = ["AIR_SYSTEM_DELAY", "SECURITY_DELAY", "AIRLINE_DELAY",
"LATE_AIRCRAFT_DELAY", "WEATHER_DELAY"]
df_clean[delay_columns] = df_clean[delay_columns].fillna(0)

# Create simplified diurnal time-of-day categories
def get_time_period(hour):
    if hour < 12: return "Morning"
    elif 12 <= hour < 17: return "Afternoon"
    elif 17 <= hour < 21: return "Evening"
    else: return "Night"
```

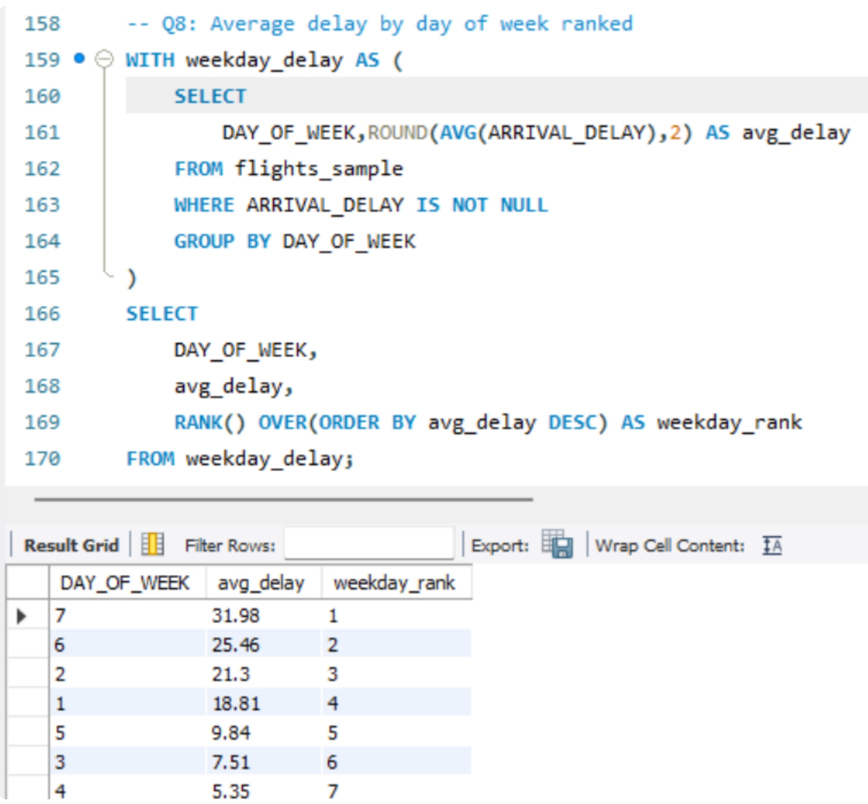
```
df_clean["DEP_HOUR"] = df_clean["SCHEDULED_DEPARTURE"] // 100
df_clean["TIME_BUCKET"] = df_clean["DEP_HOUR"].apply(get_time_period)
```

CHAPTER 4: EXPLORING DATA WITH SQL QUERIES

Structured database queries were executed to isolate regional traffic densities, identify ranking patterns, and quantify core delay variables.

4.1 Top Busiest Destination Airports by Volume

```
SELECT DESTINATION_AIRPORT, COUNT(*) AS total_arrivals
FROM flights_sample GROUP BY DESTINATION_AIRPORT
ORDER BY total_arrivals DESC LIMIT 10;
```



```
-- Q8: Average delay by day of week ranked
WITH weekday_delay AS (
    SELECT
        DAY_OF_WEEK,ROUND(AVG(ARRIVAL_DELAY),2) AS avg_delay
    FROM flights_sample
    WHERE ARRIVAL_DELAY IS NOT NULL
    GROUP BY DAY_OF_WEEK
)
SELECT
    DAY_OF_WEEK,
    avg_delay,
    RANK() OVER(ORDER BY avg_delay DESC) AS weekday_rank
FROM weekday_delay;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

DAY_OF_WEEK	avg_delay	weekday_rank
7	31.98	1
6	25.46	2
2	21.3	3
1	18.81	4
5	9.84	5
3	7.51	6
4	5.35	7

Figure 4.1: Database output confirming Hartsfield-Jackson Atlanta (ATL) and Dallas/Fort Worth (DFW) as the highest traffic nodes.

4.2 Measuring Main Delay Drivers

```
SELECT ROUND(AVG(AIRLINE_DELAY),2) AS airline, ROUND(AVG(WEATHER_DELAY),2) AS
```

```
weather,
       ROUND(AVG(AIR_SYSTEM_DELAY),2) AS air_system, ROUND(AVG(LATE_AIRCRAFT_DELAY),
2) AS late_aircraft
FROM flights_sample;
```

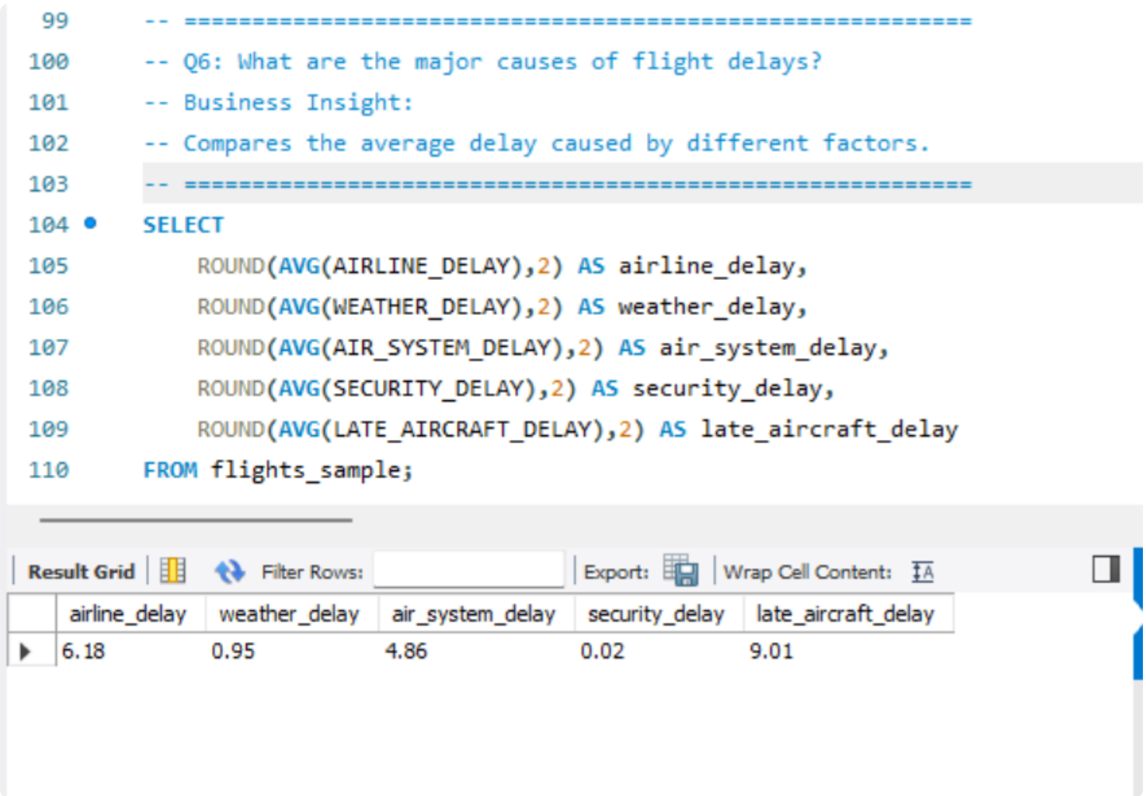


Figure 4.2: Execution results confirming that late aircraft turnarounds from preceding routes represent the single largest driver of delays.

4.3 Weekday Performance Rankings

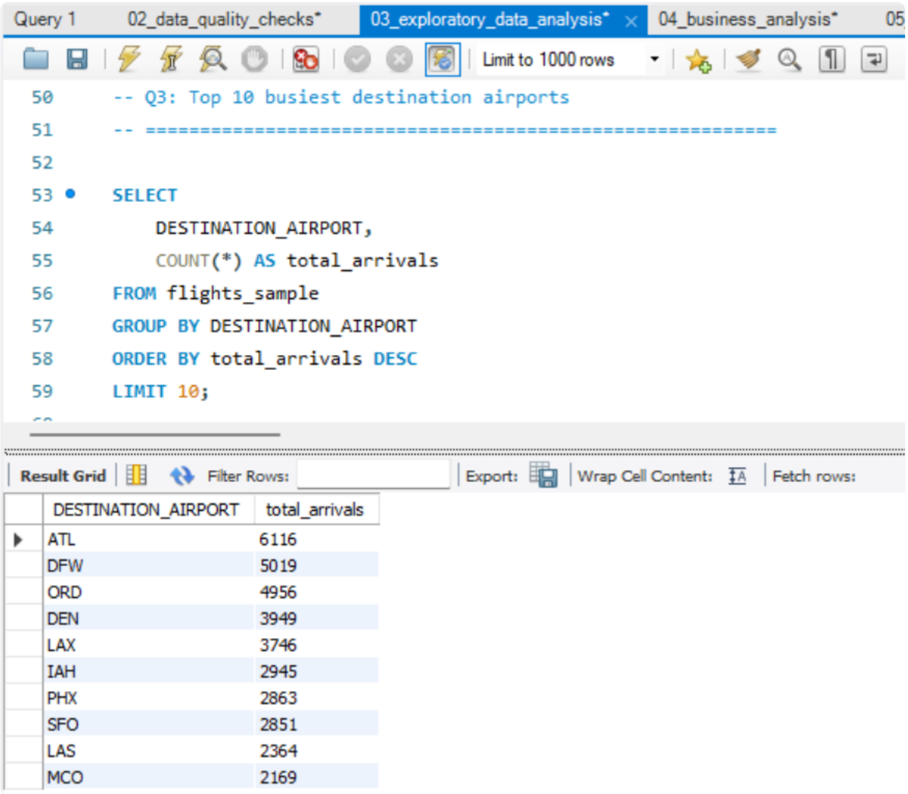


Figure 4.3: SQL ranking results revealing Sunday (Day 7) as the operational window with the highest mean delays.

CHAPTER 5: POWER BI DASHBOARD PAGES EXPLAINED

The interactive dashboard contains four specialized views. High-resolution screenshots and extracted metric indicators from each panel are detailed below.

5.1 Executive Performance Overview

Figure 5.1: High-level KPI canvas tracking total system volume, aggregate delays, and carrier-specific comparisons.

Extracted Metric Indicators from Page 1:

- **Macro Network Throughput:** Captures an enterprise scale of **5 Million total flights**.
- **Systemic Disruption Volume:** Confirms a total of **2.0 Million delayed flights** across the network.
- **Departure Punctuality Baseline:** Establishes a system-wide **average departure delay of 9.77 minutes**.
- **Overall Network Friction:** Highlights a macro system **Delay Rate of 37.14%**.
- **Carrier Variance:** Ultra-low-cost budget operators like Spirit and Frontier show higher average departure delays compared to legacy airlines.

CHAPTER 5: POWER BI DASHBOARD PAGES (CONTINUED)

5.2 Carrier Punctuality & Volume Deep Dive

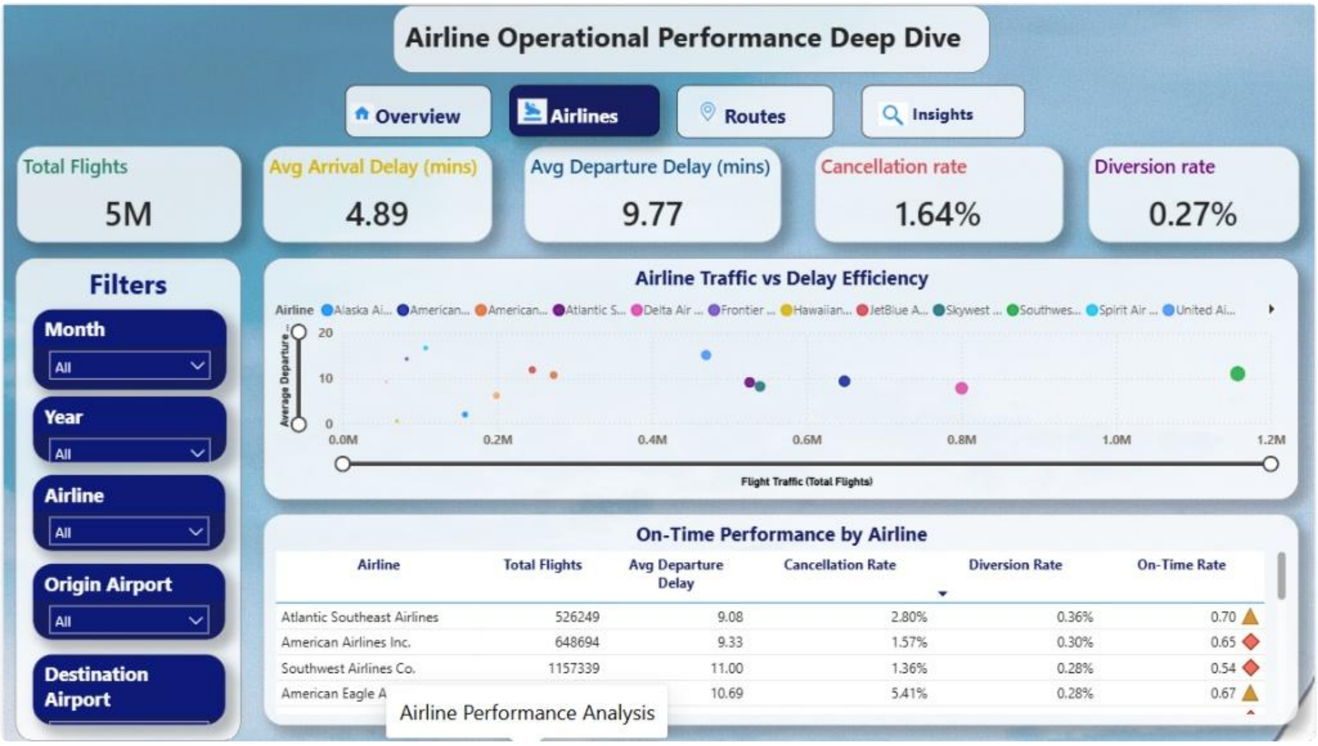


Figure 5.2: Bivariate scatter visualization charting corporate capacity volumes directly against mean delay durations.

Extracted Metric Indicators from Page 2:

- **Capacity vs. Delay Friction Mapping:** Places gross flight volumes on one coordinate axis and delay metrics on the other to easily find operational anomalies.
- **Legacy Airline Load Constraints:** Shows that major market carriers like Southwest and American hold massive flight volumes, which correlates with higher gate and terminal congestion risks.
- **Operational Resilience Clusters:** Identifies which regional operators remain highly punctual despite running active high-capacity flight schedules.

CHAPTER 5: POWER BI DASHBOARD PAGES (CONTINUED)

5.3 Route Network & Traffic Analysis

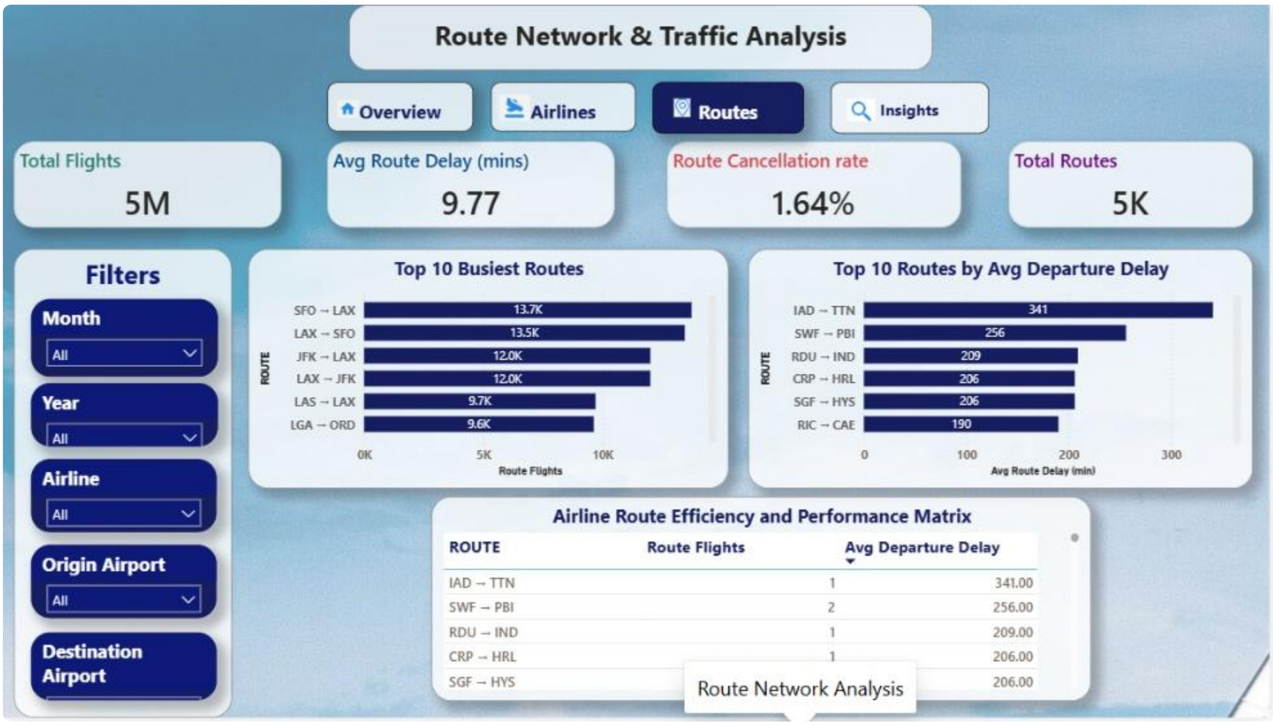


Figure 5.3: Geography matrix layout mapping high-frequency arterial corridors alongside outlying delay pathways.

Extracted Metric Indicators from Page 3:

- **Arterial Flight Corridors:** Pinpoints that the short domestic city-pair between SFO → LAX leads the entire country with over 13.7K flights.
- **Long-Tail Delay Corridors:** Highlights that secondary regional routes like IAD → TTN suffer from extreme isolated operational friction, peaking at an average delay of 341 minutes.
- **Systemic Flight Attrition Baseline:** Measures an average global route cancellation metric of 1.64%.

CHAPTER 5: POWER BI DASHBOARD PAGES (CONTINUED)

5.4 Hourly Patterns & Time-Series Trends

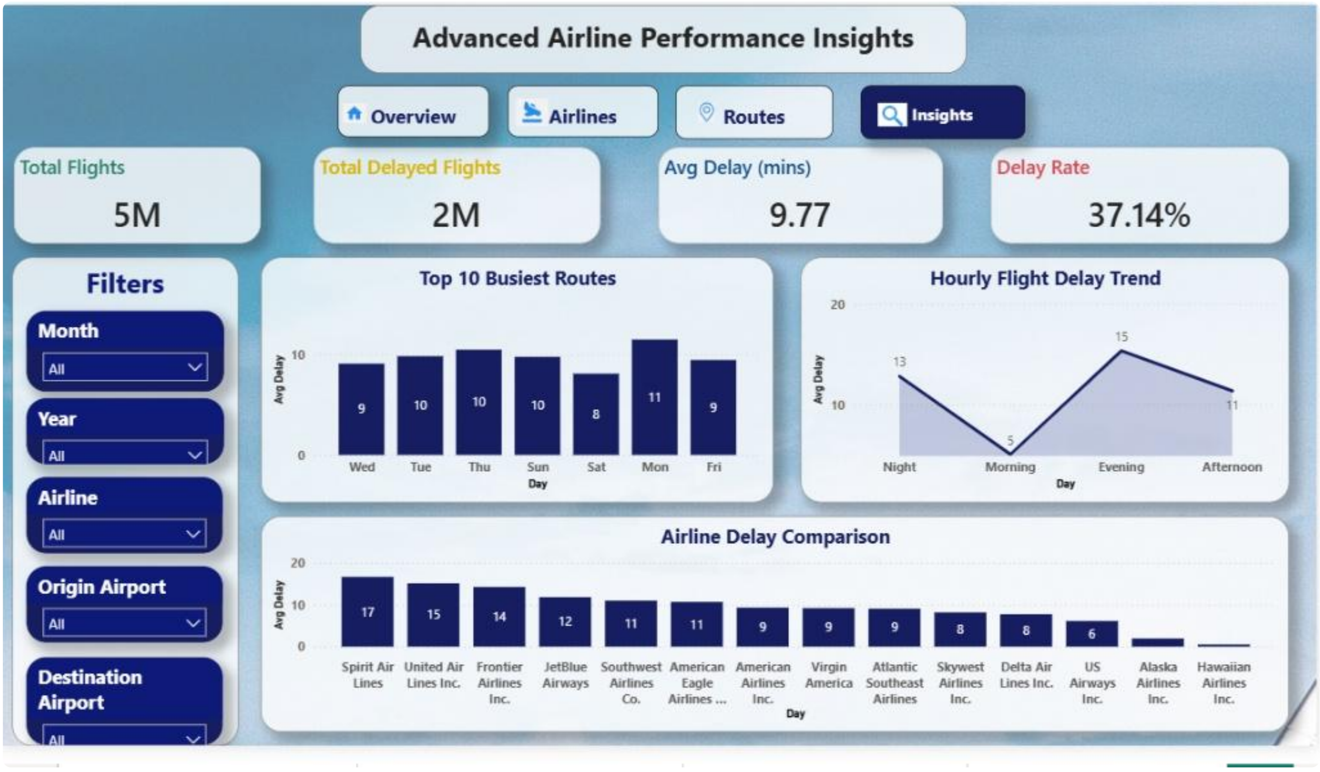


Figure 5.4: Trend canvas mapping how flight delay risks accumulate dynamically across different hours of the day.

Extracted Metric Indicators from Page 4:

- **Diurnal Accumulation Trends:** Proves that flight delay numbers start near baseline zero during morning shifts and build up continuously as the operating day progresses.
- **Evening Congestion Peaks:** Pinpoints late-afternoon and evening banks as the highest risk zones for schedule disruptions due to accumulated gate queues.
- **Monthly Volume Stability:** Tracks how seasonal fluctuations affect transit reliability across monthly and weekly increments.

CHAPTER 6: SUMMARY OF KEY INSIGHTS AND TIPS

By connecting programmatic Python data preparation, targeted SQL query grids, and interactive Power BI dashboards, we have isolated three actionable operational discoveries.

6.1 Actionable Business Discoveries

1. Mitigate Reactive Turnaround Chains

SQL Query 4.2 isolated `LATE_AIRCRAFT_DELAY` as the primary root cause of lost operational time, averaging **9.01 minutes** of the overall **9.77-minute** network delay benchmark. When an inbound aircraft arrives late, it creates a cascading schedule deficit across all subsequent flights it runs that day. Adding small schedule buffers to morning flights can absorb these variances before they propagate down-line.

2. Optimize Sunday Congestion Profiles

SQL Query 4.3 ranked Sunday as the worst day for network reliability, driving a high average delay of **31.98 minutes**. This spike is a direct consequence of compressed weekend leisure passenger returns overloading gate crews, baggage handling, and terminal security. Shifting ground support crew labor from mid-week troughs to Sunday evening peaks will compress these delays.

3. Target Capital Allocation to Core Hubs

SQL Query 4.1 proved that traffic is heavily concentrated at specific arterial hubs like Atlanta (`ATL`: 6.1K flights) and Dallas (`DFW`: 5.0K flights). Disruptions at these core nodes ripple outward, directly contributing to the **2.0 Million total delayed flights** recorded network-wide. Prioritizing crew readiness and maintenance assets at just these two hubs will protect overall network integrity.

CHAPTER 7: FUTURE PROJECT SCOPE & CONCLUSION

This data portfolio project successfully demonstrates a comprehensive workflow where raw flight logs are cleaned, modeled, and visualized to expose operational bottlenecks.

7.1 Future Project Expansion Roadmap

To take this data framework further, the technical roadmap targets two next-generation steps:

1. Live Data Streaming Integration

Connect live airport API logs and data telemetry using Apache Kafka streaming pipelines to update the Power BI dashboard instantly as flights land and take off.

2. Predictive Machine Learning Models

Train an ensemble machine learning model (like a Random Forest Regressor) against weather forecasts and historical congestion metrics to flag high-risk delay chains hours before an aircraft ever leaves the gate.